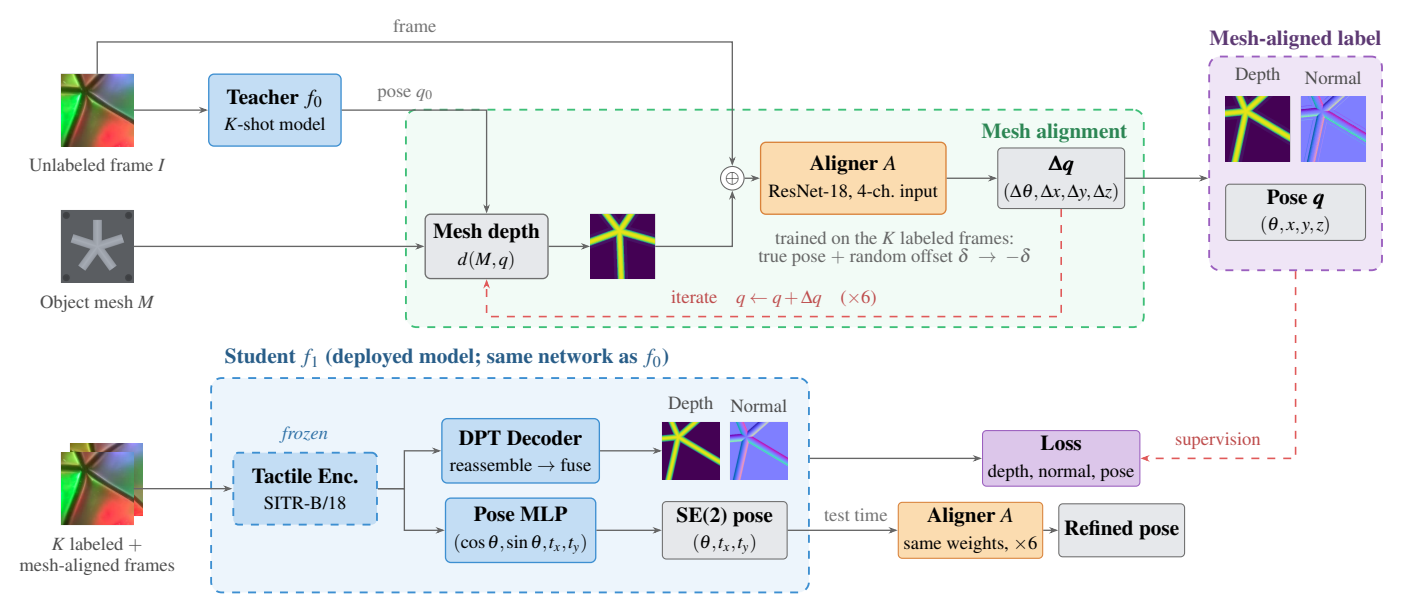

frame
Unlabeled frame I
Teacher f_0
K-shot model
pose q_0
Object mesh M
Mesh depth
$d(M,q)$
Mesh alignment
Aligner A
ResNet-18, 4-ch. input
Δq
$(\Delta\theta, \Delta x, \Delta y, \Delta z)$
trained on the K labeled frames:
true pose + random offset $\delta \rightarrow -\delta$
iterate $q \leftarrow q + \Delta q$ $(\times 6)$
Mesh-aligned label
Depth
Normal
Pose q
(θ, x, y, z)
Student f_1 (deployed model; same network as f_0)
K labeled + mesh-aligned frames
frozen
Tactile Enc.
SITR-B/18
DPT Decoder
reassemble $\rightarrow$ fuse
Pose MLP
$(\cos\theta, \sin\theta, t_x, t_y)$
Depth
Normal
SE(2) pose
(θ, t_x, t_y)
Loss
depth, normal, pose
supervision
test time
Aligner A
same weights, $\times 6$
Refined pose